

Capstone Project: Research Assistant

Project Overview

Introduction and Problem Statement

The capstone project serves as a culminating experience, allowing you to apply the knowledge and skills you have acquired throughout your studies to a real-world problem or challenge. This project will apply to a more research-focused environment where the agent will navigate extensive data collections while also using the necessary tools to generate new data. For this, the applications can be broader but more focused on a specific research problem. Examples of this include an agentic system that proposes element compositions for new alloys or airfoil designs for drag optimization. Any problem that requires collecting and creating new data using available tools falls into this category; the choice is up to you!

Project Goals and Objectives

The primary goal of this capstone is to demonstrate your ability to apply the agentic tools and concepts you have acquired throughout the course.

Concepts	Description	Success Metric
Tool Calling	Utilize an agentic architecture to provide the large language model with tool-calling abilities.	The agent can execute tool calls and use functions while understanding their outputs.
Reasoning Architectures and Chain of Thought	Apply a basic reasoning loop, such as ReAct, so the agent can interact with its current environment.	The agentic system demonstrates the ability to reason and recover from missteps.
Knowledge and Memory	Utilize a retrieval-based memory mechanism such as RAG or other proposed methods to retain knowledge between tool calls.	The agentic system can recall past task executions or use a store of knowledge, such as API documentation or other data.

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Further Reasoning	Further build up the reasoning loop using concepts such as Tree of Thought.	The system can more clearly outline the agent's reasoning architecture and thought process.
Multi-agent Coordination	Develop the agentic system to leverage parallelism and specialized agents to assist with various tasks.	Demonstration of at least one agent to execute tasks.
Safety	Ensure that the agentic system is safe to deploy and that specified guardrails are in place to prevent unintended task execution.	Statement of potential unintended actions the system may execute and the steps taken to address this.

Deliverables

The final project deliverables will include:

- **Final Report:** A comprehensive document detailing the problem, methodology, results, and conclusions.
- **Presentation:** A formal presentation summarizing the project for the faculty and peers.